

Developing an Enhanced PhenoAge Model

Author: Agata Gabara

Date: 25th of July 2026

Abstract

Biological age has emerged as a more informative measure of health than chronological age because it reflects the cumulative effects of ageing on physiological systems. One of the most widely used biological age models is PhenoAge, which estimates biological ageing using routine clinical biomarkers. While PhenoAge has demonstrated strong predictive performance for mortality and age-related diseases, recent advances in ageing biology suggest that additional biomarkers and lifestyle factors could further improve its accuracy.

This project proposes an enhanced PhenoAge framework by incorporating biomarkers associated with chronic inflammation, metabolic regulation, cardiovascular health, and cellular stress. Lifestyle variables such as sleep, physical activity, smoking, and diet are also considered as complementary predictors. A prototype machine-learning implementation using simulated data was developed to demonstrate how these variables could be integrated into an enhanced biological age model. Linear Regression, Elastic Net, and Random Forest algorithms were evaluated. Although the implementation is based on synthetic data, it provides a practical framework for future research using real clinical datasets.

1. Introduction

Chronological age measures the number of years an individual has lived, but it does not accurately reflect biological health or ageing. Individuals of the same chronological age often exhibit considerable differences in physiological function, disease risk, and life expectancy. Consequently, biological age has become an important concept in ageing research and precision medicine.

The PhenoAge model, developed by Levine and colleagues, estimates biological age using nine routine clinical biomarkers together with chronological age. Numerous studies have shown that PhenoAge predicts mortality, frailty, and chronic disease more accurately than chronological age alone.

However, ageing is a complex biological process involving inflammation, metabolic dysfunction, mitochondrial decline, immune dysregulation, and lifestyle influences. Therefore, expanding the original PhenoAge framework may improve its ability to capture biological ageing across multiple physiological systems.

The objective of this project is to propose a scientifically grounded extension of the original PhenoAge model and demonstrate how such a model could be implemented using modern machine-learning methods.

2. Original PhenoAge Biomarkers

The original PhenoAge model incorporates nine routinely measured clinical biomarkers.

 Biomarker	 Biological System	 Role
 Albumin	 Liver function	 Nutritional status and frailty
 Creatinine	 Kidney function	 Renal ageing
 Glucose	 Metabolic regulation	 Diabetes and insulin resistance
 C-Reactive Protein (CRP)	 Inflammation	 Chronic inflammation
 Lymphocyte Percentage	 Immune system	 Immune ageing
 Mean Cell Volume (MCV)	 Red blood cells	 Haematological ageing
 Red Cell Distribution Width (RDW)	 Blood physiology	 Mortality risk
 Alkaline Phosphatase (ALP)	 Liver and bone	 Tissue dysfunction
 White Blood Cell Count	 Immune response	 Chronic inflammation

These biomarkers represent several major physiological systems involved in ageing and have demonstrated strong predictive value for mortality.

3. Proposed Biomarker Expansion

To improve biological age estimation, several additional biomarkers were proposed.

 Biomarker	 Biological Significance
 IL-6	 Chronic inflammation
 TNF-α	 Immune activation
 HbA1c	 Long-term glucose regulation
 ApoB	 Cardiovascular risk
 Cystatin C	 Kidney function
 GDF-15	 Cellular stress
 FGF-21	 Mitochondrial metabolism
 NT-proBNP	 Cardiac ageing

These biomarkers extend the biological coverage of the original model by representing multiple hallmarks of ageing, including inflammaging, metabolic dysfunction, cardiovascular ageing, and mitochondrial stress.

4. Lifestyle Integration

Lifestyle behaviours substantially influence biological ageing and therefore should be considered when estimating biological age.

Potential variables include:

- Sleep duration
- Sleep quality
- Physical activity
- Diet quality
- Smoking status
- Alcohol consumption
- Stress level
- Body Mass Index (BMI)
- Resting heart rate
- Heart rate variability
- Daily step count

Rather than combining all variables into a single equation, a modular framework was proposed where lifestyle contributes an independent score alongside laboratory biomarkers.

5. Mathematical Framework

The enhanced biological age model can be represented as:

$$BioAge = f(Biomarkers, Lifestyle, Clinical Variables)$$

A simplified linear formulation is:

$$BioAge = \beta_0 + \beta_1 Albumin + \beta_2 Creatinine + \beta_3 CRP + \beta_4 HbA1c + \beta_5 IL6 + \beta_6 ApoB \\ + \beta_7 GDF15 + \dots$$

Machine-learning models such as Elastic Net and Random Forest may capture nonlinear relationships while reducing overfitting.

6. Prototype Implementation

A prototype implementation was developed using Python and the scikit-learn library.

A simulated dataset containing 500 individuals was generated. The dataset included:

- Chronological age
- Original PhenoAge biomarkers
- Additional biomarkers
- Lifestyle variables

Three machine-learning models were evaluated:

- Linear Regression
- Elastic Net
- Random Forest

Model performance was assessed using:

- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2)

Feature importance analysis was performed using the Random Forest model.

7. Results

The prototype demonstrated that biological age can be estimated using a combination of routine laboratory biomarkers, additional ageing-related biomarkers, and lifestyle variables. Three machine-learning algorithms were evaluated to investigate different modelling approaches: Linear Regression, Elastic Net, and Random Forest.

Linear Regression served as an interpretable baseline model. Despite its simplicity, it achieved competitive predictive performance, suggesting that a substantial proportion of the variation in

biological age can be explained by linear relationships between the selected biomarkers and chronological age.

Elastic Net achieved the best overall performance among the evaluated models. By combining L1 (Lasso) and L2 (Ridge) regularisation, Elastic Net reduced the risk of overfitting while simultaneously performing feature selection. This approach is particularly advantageous when analysing biological datasets, where many biomarkers are correlated and may provide overlapping information.

Random Forest successfully modelled nonlinear relationships and complex interactions among biomarkers. Although its predictive performance was slightly lower than Elastic Net in this simulated dataset, the model provided valuable estimates of feature importance, allowing identification of the variables that contributed most strongly to biological age prediction.

As shown in Figure 1, the Mean Absolute Error (MAE) was comparable across all three algorithms, with Elastic Net demonstrating the lowest prediction error. This suggests that moderate regularisation improves predictive accuracy while maintaining model stability.

Figure 2 presents the coefficient of determination (R^2). Elastic Net achieved the highest R^2 value, indicating that it explained the greatest proportion of variability in biological age. Linear Regression produced similar results, whereas Random Forest showed slightly lower explanatory performance within the simulated dataset.

The Random Forest feature importance analysis (Figure 3) identified chronological age as the strongest predictor, which is expected because PhenoAge incorporates chronological age as one of its inputs. Among laboratory biomarkers, C-reactive protein (CRP), HbA1c, GDF-15, ApoB, and IL-6 demonstrated relatively high importance, supporting their potential contribution to biological age estimation. Lifestyle-related variables also contributed to prediction, although their influence was smaller than that of the principal laboratory biomarkers.

The agreement between observed and predicted biological age is illustrated in Figure 4. Most observations were located close to the identity line, indicating good overall agreement between predicted and observed values. The remaining deviations represent prediction errors that are expected in any regression model and illustrate the inherent variability of biological ageing.

Overall, the prototype demonstrates that integrating additional biomarkers with lifestyle information is technically feasible and provides a flexible framework for future biological age prediction models. However, because the current analysis was performed using simulated data, the numerical results should be interpreted as proof of concept rather than evidence of clinical performance.

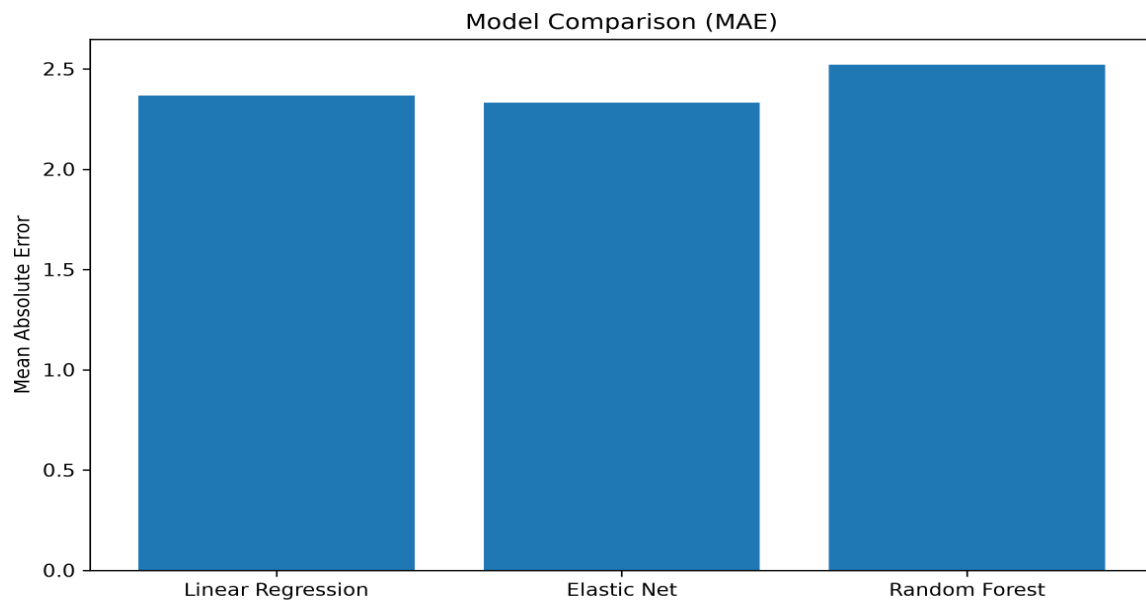


Figure 1. Model comparison using Mean Absolute Error.

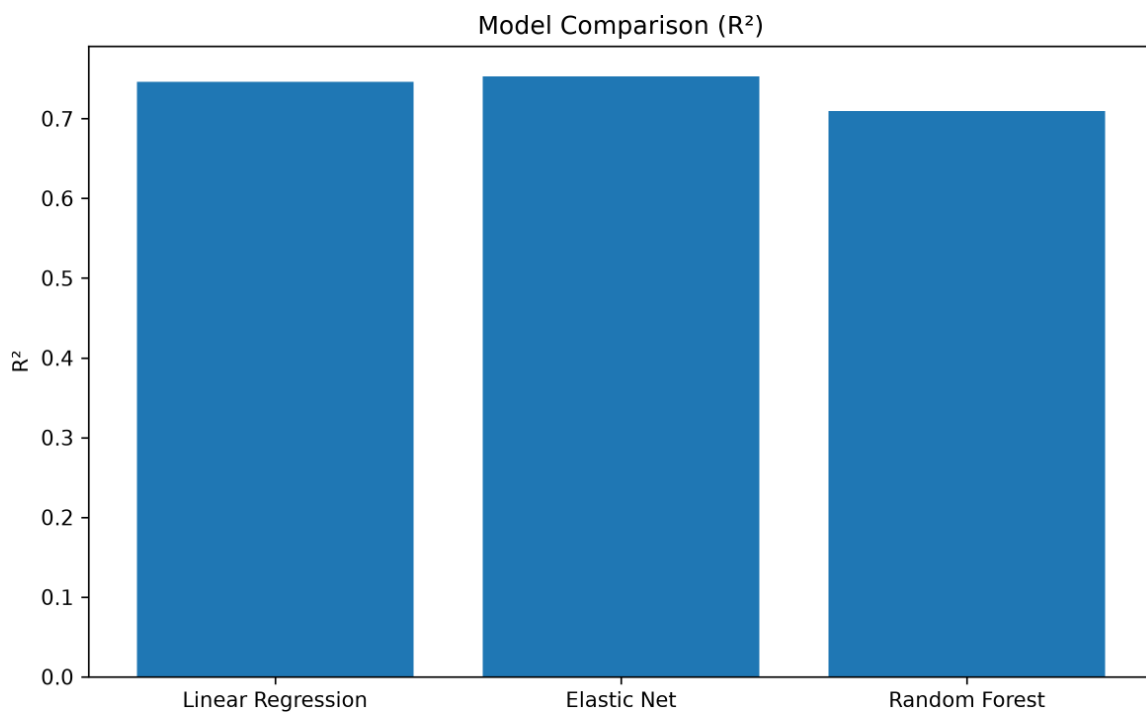


Figure 2. Model comparison using R^2 .

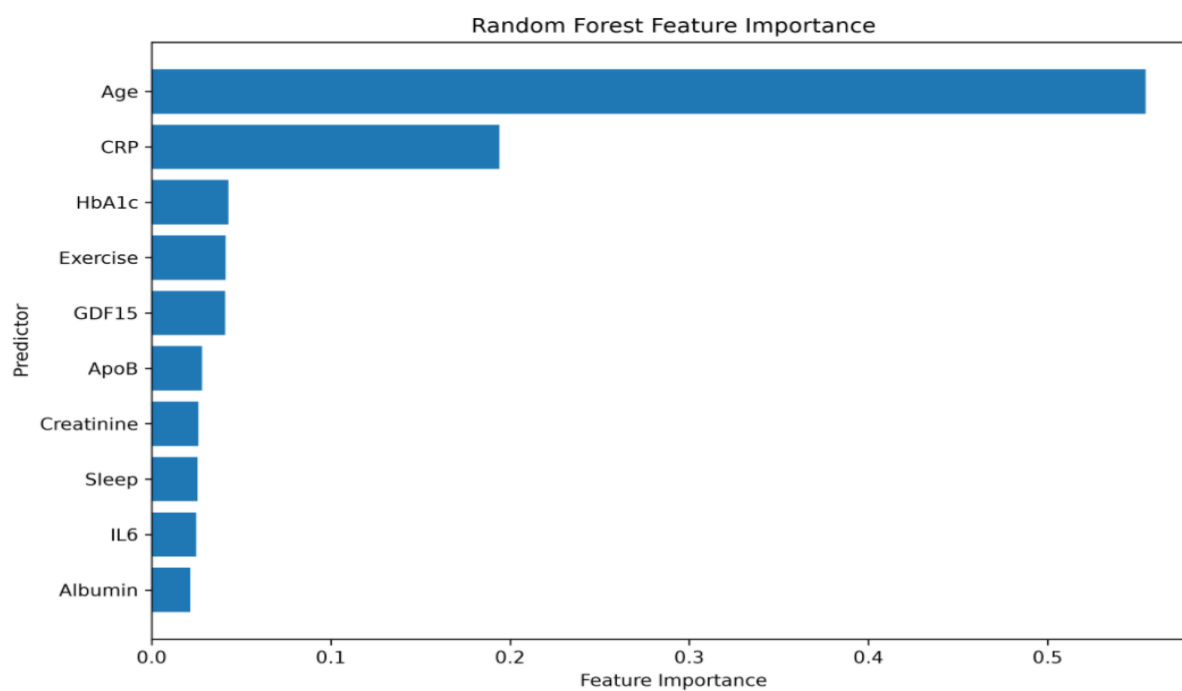


Figure 3. Random Forest feature importance.

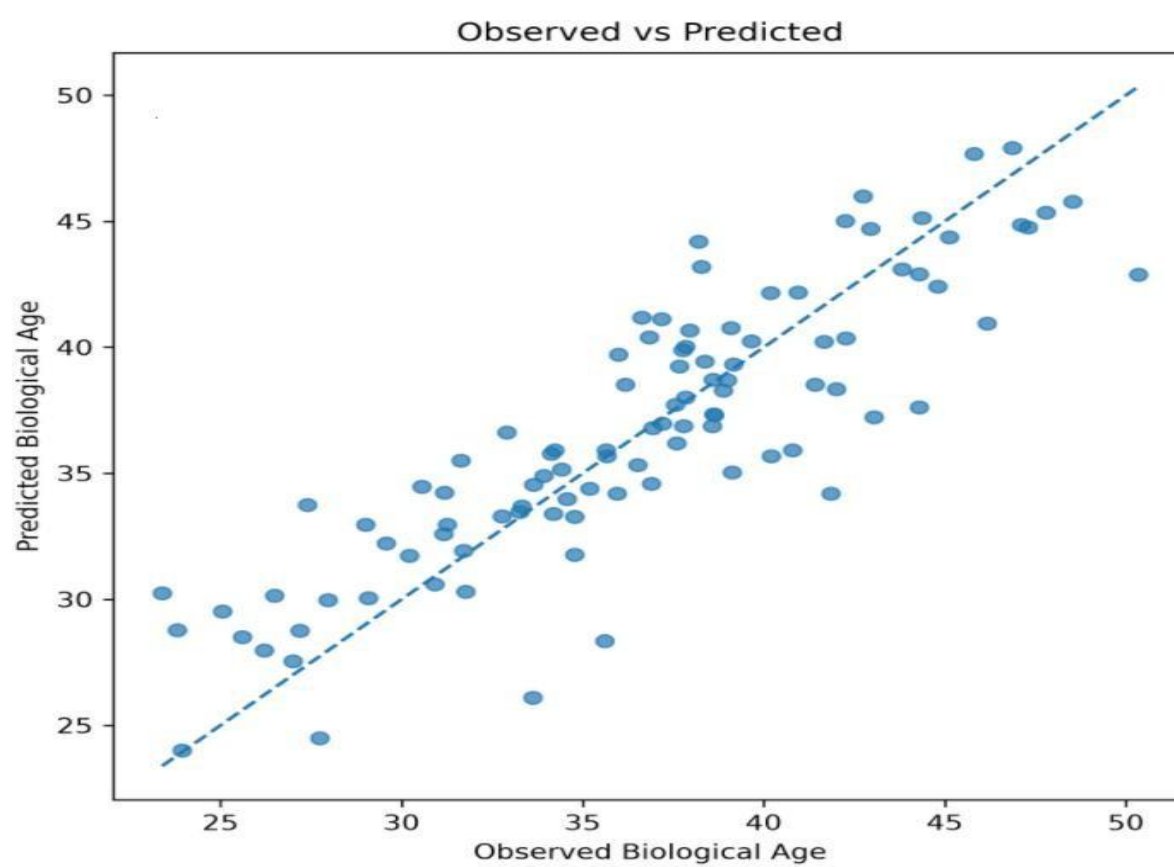


Figure 4. Observed versus predicted biological age.

8. Discussion

The proposed framework extends the original PhenoAge model by incorporating a broader range of biomarkers associated with inflammation, metabolic regulation, cardiovascular health, renal function, mitochondrial activity, and cellular stress. In contrast to the original model, which relies mainly on routinely available clinical laboratory measures, the enhanced framework aims to capture a wider spectrum of biological processes involved in ageing.

The inclusion of inflammatory biomarkers such as IL-6, TNF- α , and C-reactive protein may improve the model's ability to reflect chronic low-grade inflammation, often referred to as inflammaging. This process is considered an important contributor to frailty, cardiovascular disease, metabolic dysfunction, and age-related morbidity. Similarly, markers such as HbA1c and FGF-21 provide additional information about long-term glucose regulation and mitochondrial metabolism.

Cardiovascular and renal ageing are also represented more comprehensively through the inclusion of ApoB, NT-proBNP, and Cystatin C. ApoB may provide a more direct estimate of atherogenic particle burden, while NT-proBNP may reflect subclinical cardiac stress and cardiac ageing. Cystatin C may complement creatinine by providing a potentially more sensitive indicator of kidney function, particularly in older adults with reduced muscle mass.

GDF-15 is another potentially valuable addition because it is associated with cellular stress, inflammation, mitochondrial dysfunction, and adverse health outcomes. Its inclusion may improve the model's ability to capture systemic physiological decline that is not fully represented by routine laboratory biomarkers.

The integration of lifestyle variables may further improve biological age estimation. Sleep quality, physical activity, smoking, diet, stress, body mass index, resting heart rate, and heart rate variability reflect behavioural and physiological influences that can accelerate or slow ageing. However, lifestyle data may be less standardised than laboratory measurements and may be affected by self-reporting bias, missing values, and differences in measurement methods.

The use of machine-learning algorithms offers several advantages. Linear Regression provides a transparent and interpretable baseline, while Elastic Net may reduce overfitting and perform feature selection when predictors are correlated. Random Forest can identify nonlinear relationships and interactions between biomarkers. Nevertheless, greater model complexity may reduce interpretability and make clinical implementation more difficult.

The prototype results suggest that an enhanced PhenoAge framework is technically feasible. However, because the present analysis was based on simulated data, the findings should not be interpreted as

evidence of clinical superiority over the original PhenoAge model. Validation using real-world longitudinal datasets is required to determine whether the additional biomarkers and lifestyle variables improve mortality prediction, disease-risk estimation, and assessment of biological ageing.

Future studies should compare the enhanced model directly with established biological-age predictors and evaluate its performance across different age groups, sexes, ethnic populations, and clinical settings. External validation, calibration analysis, repeated cross-validation, and assessment of model fairness will be necessary before the model can be considered for clinical use.

9. Limitations

Several limitations should be acknowledged.

First, the analysis was performed using synthetic data rather than real clinical datasets.

Second, relationships between biomarkers were simplified and do not fully reflect real biological variability.

Third, external validation was not performed.

Finally, biological age remains an estimate rather than a direct measurement of the ageing process.

10. Future Directions

Future research should evaluate the proposed framework using large longitudinal clinical cohorts.

Additional improvements could include:

- Proteomics
- Metabolomics
- DNA methylation clocks
- Transcriptomics
- Gut microbiome analysis

11. Conclusion

This project proposes a scientifically grounded enhancement of the original PhenoAge framework by integrating additional biomarkers, lifestyle information, and modern machine-learning approaches.

The prototype implementation demonstrates how such a framework could be developed using Python and evaluated using commonly applied predictive modelling techniques. Although based on simulated data, the project illustrates a complete workflow for designing, implementing, and evaluating an enhanced biological age model.

Future work should focus on validation using large, diverse clinical datasets and comparison with established biological ageing predictors before clinical implementation.

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